

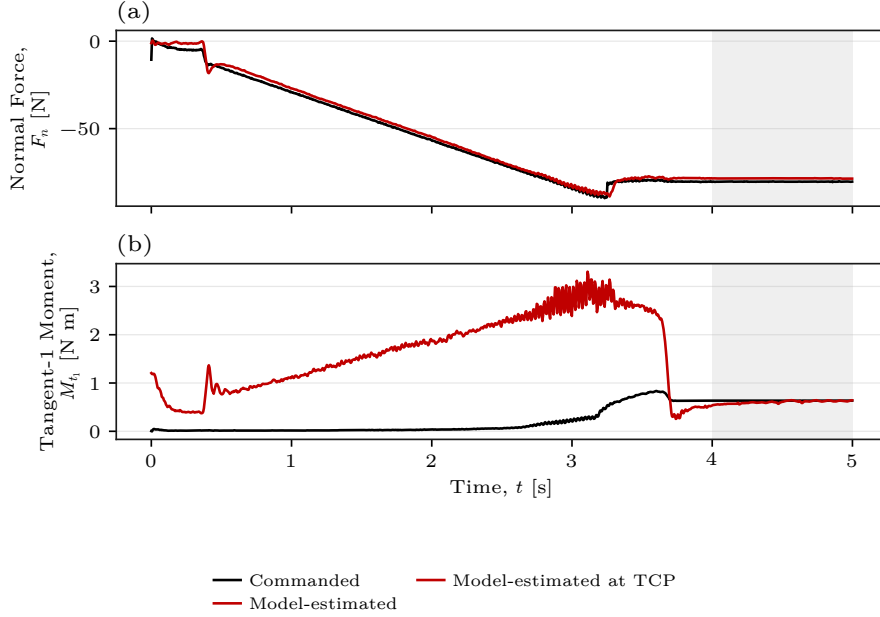
Contact-Establishment Wrench Trace, $r_c = 0$

Used impedance. The following gains were applied in the surface basis $[t_1, t_2, n_s]$:

$$\begin{aligned} K_p &= [2000, 2000, 350] \text{ N/m}, & D_p &= [231.2, 177.9, 112.1] \text{ N s/m}, \\ K_R &= [5, 5, 50] \text{ N m/rad}, & D_R &= [1.5, 1.9, 1.2] \text{ N m s/rad}. \end{aligned}$$

The centre of compliance was placed at the TCP, with $r_c = p_c - p_{\text{TCP}} = 0$. The first tangent t_1 is the normalised projection of the base-frame x_0 -direction onto the configured surface plane:

$$t_1 = \frac{e_{x_0} - (n_s \cdot e_{x_0}) \cdot n_s}{\|e_{x_0} - (n_s \cdot e_{x_0}) \cdot n_s\|} = (0.999851, -0.000477, -0.017238)^\top.$$



The curves show one five-second Contact Establishment trace. The grey area from 4 to 5 s is the stationary interval used for the reported means. Both plotted signals are absolute quantities resolved on the same surface axes. The model estimate follows the robot-to-surface action convention. The physical force acting on the robot is denoted throughout by

$$O_{f_{\text{contact}}} \equiv O_{f_{\text{surface} \rightarrow \text{robot}}} = -O_{f_{\text{robot} \rightarrow \text{surface}}}.$$

Normal-force comparison, F_n

The mean desired and measured TCP positions in the grey interval were

$$\begin{aligned} p_d &= (0.511749, -0.005127, -0.210071)^\top \text{ m}, \\ p_{\text{TCP}} &= (0.513909, 0.005024, 0.019199)^\top \text{ m}. \end{aligned}$$

Their normal error gives

$$\begin{aligned} e_n &= n_s^\top \cdot (p_d - p_{\text{TCP}}) = -0.229466 \text{ m}, \\ F_{n,\text{qs}} &= K_{p,n} \cdot e_n = 350 \cdot (-0.229466) = -80.313 \text{ N}, \\ F_{n,\text{cmd}} &= -80.312 \text{ N}, & F_{n,\text{est}} &= -78.451 \text{ N}. \end{aligned}$$

The estimate differs from the command by 1.861 N, or 2.32%. The quasi-static value and the full command differ by 0.001 N, since the mean normal speed was nearly zero. The corresponding physical reaction on the robot is

$$F_{n,\text{contact}} = n_s^\top \cdot f_{\text{contact}} = -F_{n,\text{est}} = +78.451 \text{ N}.$$

Moment comparison about t_1

The centre of compliance is at the TCP. It therefore introduces no virtual force-to-moment coupling:

$$m_{\text{CoC}} = r_c \times f = 0.$$

The physical contact quantities shown in Figure 1 are

$$\begin{aligned} r_{\text{contact}/\text{TCP}} &= p_{\text{contact}} - p_{\text{TCP}}, \\ O_{m_{\text{contact},\text{TCP}}} &= r_{\text{contact}/\text{TCP}} \times O_{f_{\text{contact}}}, \\ M_{t_1,\text{contact}} &= t_1^\top \cdot O_{m_{\text{contact},\text{TCP}}}. \end{aligned}$$

The pressure centre was not measured, so no contact lever is assigned a measured value.

In the grey interval, the measured orientation error and angular speed give

$$\begin{aligned} e_{R,t_1} &= 0.126923 \text{ rad} = 7.272^\circ, \\ \omega_{t_1} &= -0.0000185 \text{ rad/s}, \\ M_{t_1,\text{cmd}} &= K_{R,t_1} \cdot e_{R,t_1} - D_{R,t_1} \cdot \omega_{t_1} \\ &= 5 \cdot 0.126923 - 1.5 \cdot (-0.0000185) \\ &= +0.635 \text{ N m}. \end{aligned}$$

The quasi-static spring value is also $+0.635 \text{ N m}$. Force and moment equilibrium then require the opposing surface-on-robot moment

$$\begin{aligned} M_{t_1,\text{contact}} + M_{t_1,\text{cmd}} &= 0, \\ M_{t_1,\text{contact}} &= -0.635 \text{ N m}. \end{aligned}$$

With the model-estimated force magnitude, this corresponds to the quasi-static equivalent lever

$$r_{\perp,\text{qs}} = \frac{|M_{t_1,\text{contact}}|}{|F_{n,\text{est}}|} = \frac{0.635}{78.451} = 0.00809 \text{ m} = 8.09 \text{ mm}.$$

This is an equilibrium equivalent, not a measured contact location.

The measured robot state confirmed ${}^{EE}T_K = I$, so stiffness frame K coincides with the controller TCP. The base-expressed spatial wrench is therefore shifted to the TCP by

$$\begin{aligned} {}^Om_{\text{TCP},\text{est}} &= {}^Om_{O,\text{est}} - p_{\text{TCP}} \times {}^Of_{\text{est}}, \\ M_{t_1,\text{est}} &= t_1^\top \cdot {}^Om_{\text{TCP},\text{est}} = +0.604 \text{ N m}. \end{aligned}$$

The estimate uses the opposite action side to $M_{t_1,\text{contact}}$. It is therefore compared with $-M_{t_1,\text{contact}} = +0.635 \text{ N m}$, as shown in Figure 1. Its difference from the command is 0.031 N m , or 4.87% . Subtracting the non-zero clearance estimate would instead give -0.601 N m ; that change signal is not compared with the absolute command and is not plotted here.

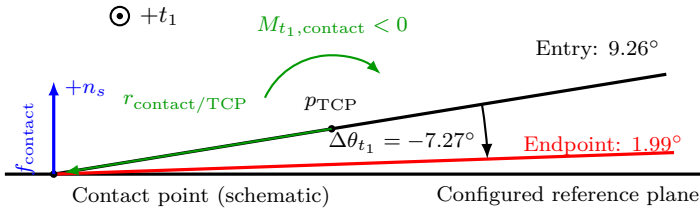
Orientation change

The measured component about t_1 fell from 9.26° at contact entry to 1.99° at the endpoint. Thus,

$$\Delta\theta_{t_1} = 1.99^\circ - 9.26^\circ = -7.27^\circ.$$

The off-centre contact rotates the tool towards the plane. The positive rotational command develops in the opposing direction and retains the entry orientation.

(a) Physical action on the robot



(b) Stationary-interval moment means

Grey interval: 4–5 s

Quasi-static balance

$$M_{t_1,\text{contact}} = -0.635 \text{ N m}$$

$$M_{t_1,\text{cmd}} = +0.635 \text{ N m}$$

$$M_{t_1,\text{contact}} + M_{t_1,\text{cmd}} = 0$$

Command–estimate comparison

$$M_{t_1,\text{cmd}} = +0.635 \text{ N m}$$

$$M_{t_1,\text{est}} = +0.604 \text{ N m}$$

Figure 1: Contact response and endpoint moment comparison.

Normal-force summary. The quasi-static, commanded and model-estimated endpoint values are -80.313 , -80.312 and -78.451 N . The estimate differs from the command by 2.32% .

Moment summary. The quasi-static, commanded and model-estimated endpoint values are $+0.635$, $+0.635$ and $+0.604 \text{ N m}$. The estimate differs from the command by 4.87% . This single Contact Establishment trace is consistent in sign and magnitude for both components.

T-Mode Quasi-Static Consistency Test

The test uses the decoupled setup impedance in surface coordinates, with $r_c = 0$. The commanded components are

$$\begin{aligned} F_{n,\text{cmd}} &= K_{p,n} \cdot e_n + D_{p,n} \cdot \dot{e}_n, \\ M_{t_1,\text{cmd}} &= K_{R,t_1} \cdot e_{R,t_1} + D_{R,t_1} \cdot \dot{e}_{R,t_1}. \end{aligned}$$

During each stationary hold, $\dot{e}_n \simeq 0$ and $\dot{e}_{R,t_1} \simeq 0$. The quasi-static predictions therefore reduce to

$$F_{n,\text{qs}} = K_{p,n} \cdot e_n, \quad M_{t_1,\text{qs}} = K_{R,t_1} \cdot e_{R,t_1}.$$

The consistency measurement used

$$K_{p,n} = 1000 \text{ N/m}, \quad D_{p,n} = 175 \text{ N s/m}, \quad K_{R,t_1} = 15 \text{ N m/rad}, \quad D_{R,t_1} = 10.01 \text{ N m s/rad}.$$

The accepted nominal checks were approximately 20 mm and 5° . At exactly 20 mm, the force prediction is

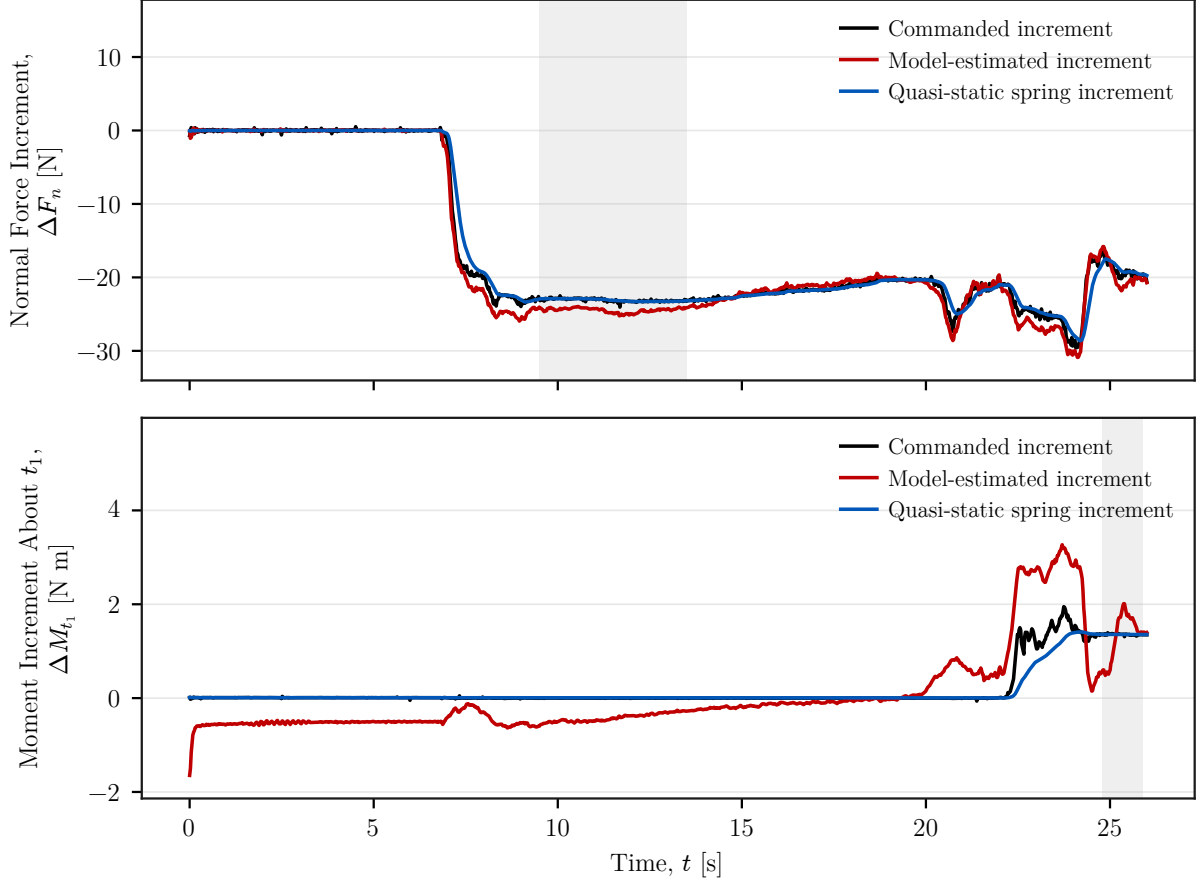
$$|F_{n,\text{qs}}| = 1000 \cdot 0.020 = 20.0 \text{ N},$$

and at exactly $5^\circ = 0.087266 \text{ rad}$, the moment prediction is

$$|M_{t_1,\text{qs}}| = 15 \cdot 0.087266 = 1.309 \text{ N m}.$$

T-Mode Consistency Results

The comparison contains one trial. The grey band in the upper F_n panel marks 9.5–13.5 s, during which the normal displacement was held stationary. The grey band in the lower M_{t_1} panel marks 24.8–25.9 s, during which the rotation about t_1 was held stationary. Only the samples inside the corresponding grey band are used for each plateau mean and consistency check.



Component	Achieved	Quasi-static	Commanded	Estimated	Status
F_n	-23.09 mm	-23.089	-23.093	-24.461	consistent
M_{t_1}	5.196 deg	1.360	1.358	1.346	consistent

Normal force F_n .

$$F_{n,qs} = K_{p,n} \cdot e_n = 1000 \cdot (-0.023089) = -23.089 \text{ N.}$$

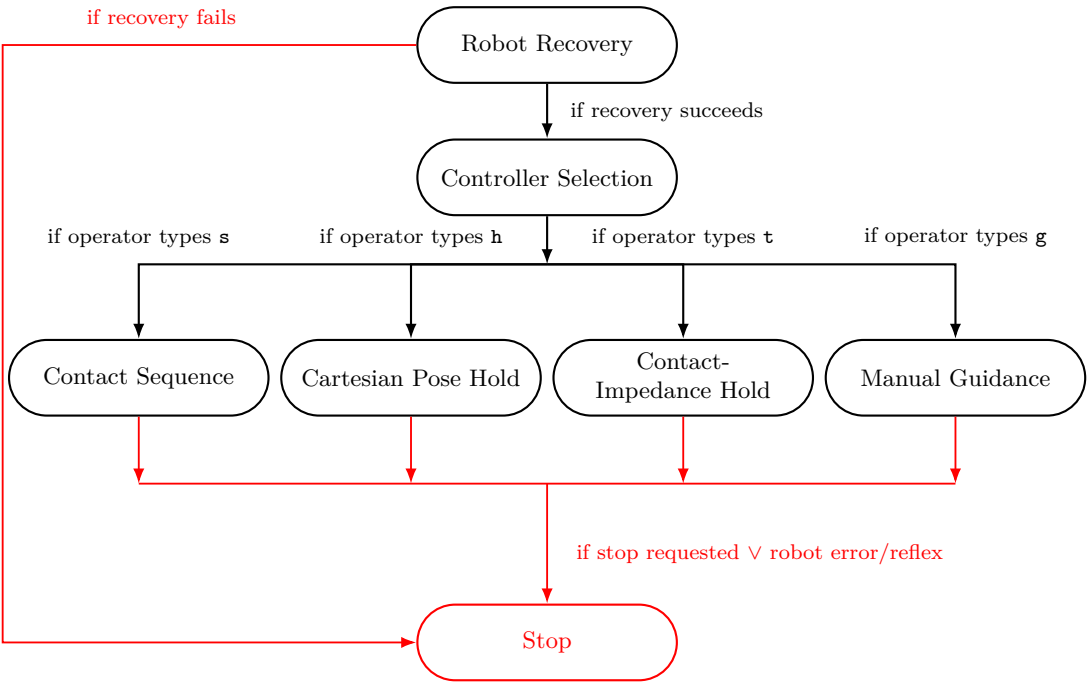
The commanded mean is -23.093 N. The quasi-static difference is 0.004 N (0.02%). The model-estimated mean is -24.461 N, a difference of 1.368 N (5.92%).

Moment M_{t_1} .

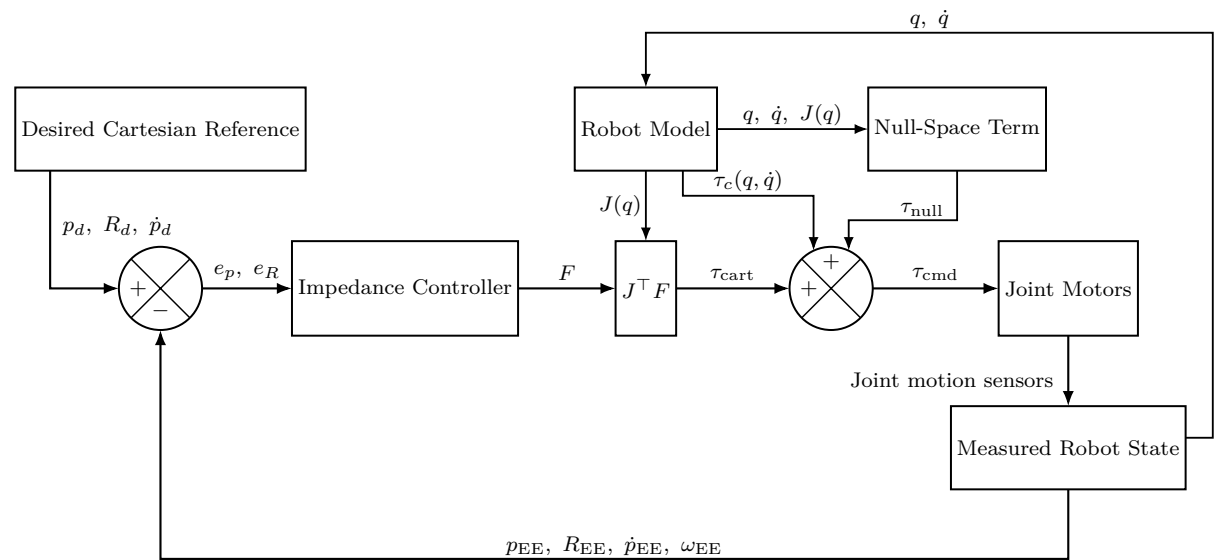
$$M_{t_1,qs} = K_{R,t_1} \cdot e_{R,t_1} = 15 \cdot (5.196\pi/180) = 1.360 \text{ N m.}$$

The commanded mean is 1.358 N m. The quasi-static difference is 0.003 N m (0.20%). The model-estimated mean is 1.346 N m, a difference of 0.012 N m (0.87%).

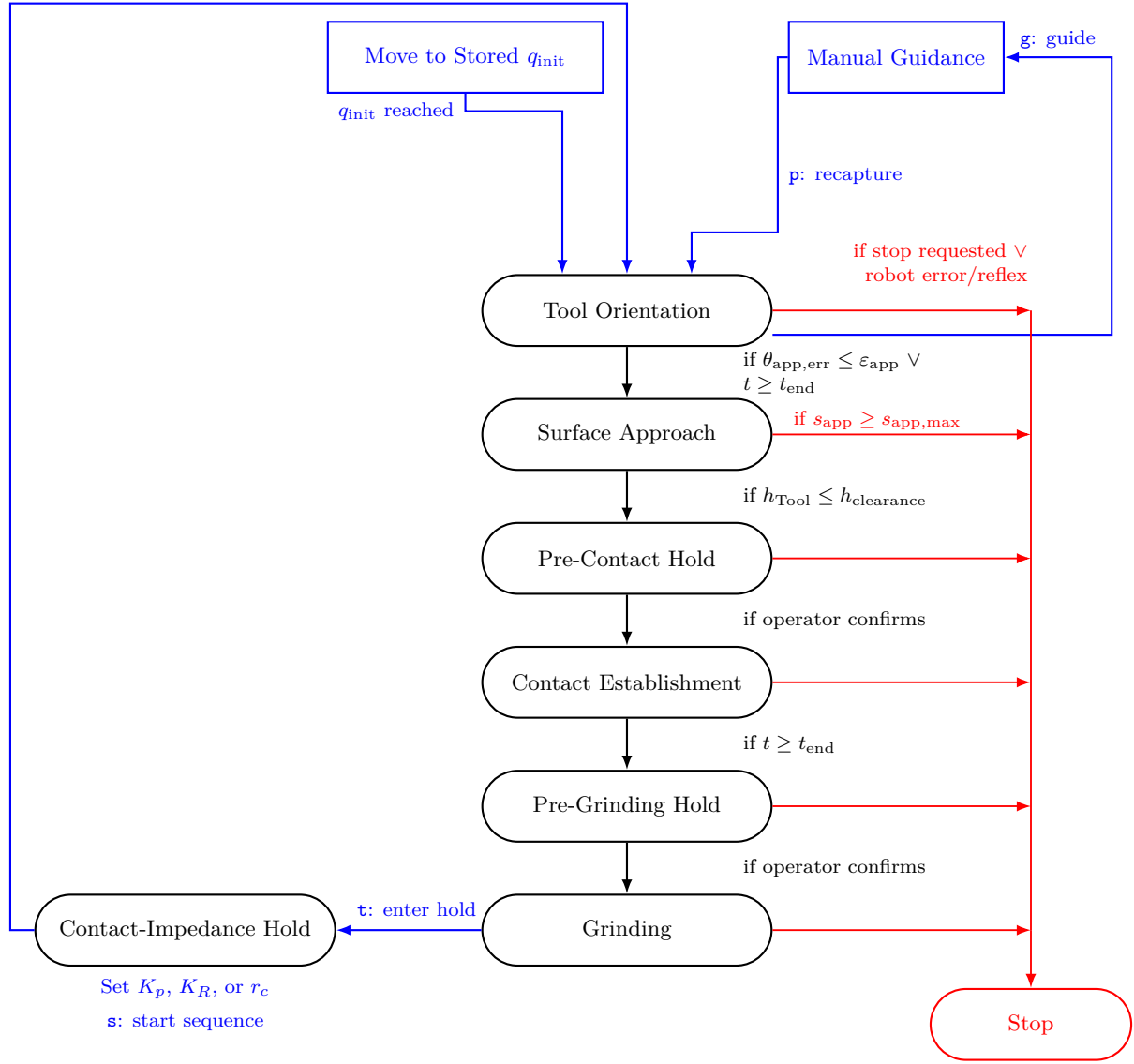
Controller State Flow



Cartesian Impedance Block Diagram



Surface-Contact Sequence



Cartesian Pose Hold

